

Erdem Bıyık

<https://ebiyik.github.io>

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CURRENT POSITION

University of Southern California

- Assistant Professor of Computer Science

Assistant Professor of Electrical and Computer Engineering - Systems (by courtesy)

Los Angeles, CA, USA

Aug 2023 – present

Jan 2024 – present

EDUCATION

Stanford University

- Doctor of Philosophy in Electrical Engineering; GPA: 4.07/4.00

Advisor: Prof. Dorsa Sadigh

Stanford, CA, USA

Sep 2017 – Jun 2022

Stanford University

- Master of Science in Electrical Engineering; GPA: 4.08/4.00

Stanford, CA, USA

Sep 2017 – Apr 2019

Bilkent University

- Bachelor of Science in Electrical and Electronics Engineering; GPA: 4.00/4.00

Rank: 1/170 based on GPA

Ankara, Turkey

Aug 2012 – Jun 2017

National University of Singapore

- Exchange Student in Electrical and Computer Engineering

Singapore

Aug 2015 – Dec 2015

WORK EXPERIENCE

UC Berkeley, Center for Human-Compatible Artificial Intelligence (CHAI)

Berkeley, CA, USA

- Postdoctoral Researcher

Jul 2022 - Jul 2023

- **Reward Learning:** Working on various active reward learning projects for robotics under the supervision of Prof. Stuart Russell and Prof. Anca Dragan
- **Reinforcement Learning for Deep Brain Stimulation:** Working on the applications of reinforcement learning algorithms for deep brain stimulation under the supervision of Prof. Anca Dragan

Google Research

Mountain View, CA, USA

- Research Intern

Jun 2021 - Sep 2021

- **Recommender Systems:** Working on active preference elicitation for recommender systems under the supervision of Dr. Yinlam Chow, Dr. Mohammad Ghavamzadeh, and Prof. Craig Boutilier
- **Preference-based Reinforcement Learning:** Working on relaxing initial state assumptions in preference-based reinforcement learning under the supervision of Dr. Yinlam Chow and Dr. Mohammad Ghavamzadeh

National Magnetic Resonance Research Center (UMRAM)

Ankara, Turkey

- Undergraduate Researcher & Research Intern

Apr 2016 - Sep 2017

- **csMRI:** Developing compressed sensing methods for accelerated MRI under the supervision of Prof. Tolga Çukur
- **SSFP Imaging:** Developing coil compression methods and artifact suppression techniques for balanced SSFP under the supervision of Prof. Tolga Çukur

ASELSAN

Ankara, Turkey

- Intern (2015), Research Engineer (2017)

Jun 2015 - Jul 2015, Apr 2017 - Aug 2017

- **Military Communications (2015):** Developing a C++ program to decode and encode data in MIL-STD-3014 protocol with an easy-to-use interface
- **Algorithms Design (2015):** Designing algorithms to solve composite launch acceptability region problem

- **SSFP Imaging (2017):** R&D projects related to field inhomogeneity, banding profile estimation, coil compression and banding suppression in bSSFP MRI under the supervision of Dr. Aykut Koç

- **École polytechnique fédérale de Lausanne (EPFL)** Lausanne, Switzerland
Research Intern via Summer@EPFL Program *Jun 2016 - Sep 2016*
 - **Approximate Message Passing:** Research about approximate message passing algorithms as an intern in the Information Processing Group under the supervision of Prof. Rüdiger Urbanke
- **Anadolu Agency** Ankara, Turkey
Intern *Jun 2014 - Jul 2014*
 - **Image Processing:** Developing a Java, C++ and MySQL based program that uses image registration, matching and comparison techniques to detect copyright infringements

TEACHING EXPERIENCE

- **University of Southern California** Los Angeles, CA
Instructor *Fall 2023, Spring 2024*
 - **CSCI445L Introduction to Robotics (Spring 2024):** Teaching the course
 - **CSCI699 Robot Learning (Fall 2023):** Designing and teaching the course
- **Stanford University** Stanford, CA
Teaching Assistant *Winter 2020, Winter 2021*
 - **CS237B / EE260B / AA174B / AA274B Principles of Robot Autonomy II:** Holding office hours and sections, preparing and grading homeworks and exams
Awarded “outstanding course assistant” by the CS department (top 5%) in Winter 2021
- **Bilkent University** Ankara, Turkey
Teaching Assistant *Spring 2014, Fall 2016*
 - **CS114 Introduction to Programming for Engineers (2014):** Teaching in the weekly tutorials & recitations
 - **EEE211 Analog Electronics (2016):** Teaching, interviewing and grading students in the laboratory sessions
- **Stanford University** Stanford, CA
Guest Lecturer *Winter 2022*
 - **CS333 Safe and Interactive Robotics:** Teaching “learning from human preferences”

MENTORING

- **Current Ph.D. Students:** Ayush Jain (co-advised with Joseph Lim), Jesse Zhang (co-advised with Jesse Thomason and Joseph Lim), Anthony Liang (co-advised with Stephen Tu), Pavel Czempin, Yigit Korkmaz, Yutai Zhou
- **Current M.Sc. Students:** Xinhui Li, Dhanush Kumar Varma Penmetsa, Zhaojing Yang
- **Current Undergraduate Students:** Jaiv Doshi, Miru Jun
- **Ph.D. Alumni**
 - Sumedh Sontakke (2024, co-advised with Laurent Itti). Next position: Amazon Robotics.
- **M.Sc. Alumni**
 - Thomas Reeves (2024). Next position: Ph.D. in Computer Science at USC.

HONORS & AWARDS

- **Stanford CS Outstanding Course Assistant:** Awarded to the top 5% of CAs in the department. Awarded for the “Principles of Robot Autonomy II” course in Winter 2021
- **Qualcomm Innovation Fellowship 2020 North America:** Finalist (as a group of two PhD students)
- **HRI 2020 Honorable Mention:** Awarded for the paper “When Humans Aren’t Optimal: Robots that Collaborate with Risk-Aware Humans”
- **Qualcomm Innovation Fellowship 2019 North America:** Finalist (as a group of two PhD students)
- **Stanford University James D. Plummer Graduate Fellowship (2017-2018):** Full tuition waiver & stipend during the first year of PhD program
- **Bilkent University Electrical and Electronics Engineering Department, Graduation Awards (2017):** Academic Excellence, Research Excellence, Voluntary Professional Activities, Social Awareness and Activities
- **Scholarship of the Turkish Prime Ministry (2012-2017):** Awarded monthly stipend during the BSc program (given to those who rank in first 100 among 1.8 million students in nationwide university entrance exam)
- **Bilkent University Comprehensive Scholarship (2012-2017):** Full tuition waiver & stipend during the BSc program
- **IEEEExtreme Programming Competitions:** 2nd in Turkey, 78th among all participants, 2015. 1st in Turkey, 73rd among all participants, 2014. 1st in Turkey, 100th among all participants, 2013 (All as a group of three students)
- **Turkish Intelligence Foundation (TZV) Marathon:** Ranked twice in top 25 and once 6th, 2012-2014
- **İşbank Golden Youth Award (2012):** Granted for outstanding performance in the nationwide university entrance exam
- **Nationwide University Entrance Exam (LYS):** Ranked 13th among 1.8 million students in Turkey, 2012

INVITED TALKS & POSTER PRESENTATIONS

- Data- and Time-Efficient Learning of Human Preferences
 - 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2024)
- Robot Learning with Minimal Human Feedback
 - University of Illinois Urbana-Champaign, Human-Centered Autonomy Lab (2024)
- Optimizing Robot Behavior via Comparative Language Feedback
 - Human-Interactive Robot Learning (Workshop at HRI 2024 – poster)
- Data-Efficient Learning from Human Feedback and Large Pretrained Models for Robotics
 - Arizona State University, School of Computing and Augmented Intelligence (2023)
- Open Problems in Reinforcement Learning from Human Feedback and Potential Solutions for Data-Efficiency
 - University of Amsterdam, The Amsterdam Machine Learning Lab (2023)
- Efficient Robot Learning via Interaction with Humans
 - Southern California Robotics (SCR) Symposium 2023
- ViSaRL: Visual Reinforcement Learning Guided by Human Saliency
 - 7th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2023 – poster)
- Learning Preferences for Interactive Autonomy
 - Middle East Technical University, Robotics and AI Technologies Application and Research Center (2022)
 - Sonoma State University, Engineering Colloquium (2022)
 - Cornell University, Robotics Seminar (2022)
 - University of Wisconsin-Madison, Computer Science, Department Seminar (2022)

- University of Illinois Urbana-Champaign, Computer Science, Department Seminar (2022)
- University of Michigan, Robotics Institute, Department Seminar (2022)
- University of Southern California, Computer Science, Department Seminar (2022)
- Imperial College London, Department of Aeronautics, Aerodynamics & Control Seminar (2022)
- Carnegie Mellon University, Robotics Institute, Department Seminar (2022)
- UCLA, Electrical and Computer Engineering, Department Seminar (2022)
- Bilkent University, Electrical and Electronics Engineering, Graduate Seminar (2021)
- UC Berkeley, Center for Human-Compatible Artificial Intelligence (CHAI), Beneficial AI Seminar (2021)
- Sabancı University, Computer Science & Engineering, Department Seminar (2021)
- Koç University, AI Meetings (2021)
- Bay Area Robotics Symposium (BARS) 2021 (poster)
- Virginia Tech, Mechanical Engineering, Department Seminar (2021)
- Caltech Yue Lab (2021)
- UT Austin, Personal Autonomous Robotics Lab (2021)
- UC Berkeley, InterACT Lab (2021)
- Learning Multimodal Rewards from Rankings
 - 6th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2022)
 - CoRL 2021
- Learning from Humans for Adaptive Interaction
 - HRI Pioneers 2022
- APReL: A Library for Active Preference-based Reward Learning Algorithms
 - Human-Interactive Robot Learning (HIRL) (Workshop at HRI 2022)
 - HRI 2022
 - AI-HRI 2021 at AAAI Fall Symposium Series
- Partner-Aware Algorithms in Decentralized Cooperative Bandit Teams
 - AAAI 2022
 - AI-HRI 2021 at AAAI Fall Symposium Series
- The Role of Representations in Human-Aware Learning and Control (*with Dorsa Sadigh*)
 - “Aware-Learning: How to Benefit from Priors” Workshop @ CDC 2021
- Interactive Robotics through the Lens of Learning (*with Dorsa Sadigh*)
 - NCCR (The National Centre of Competence in Research, Switzerland) Automation Seminar, 2021
- Leveraging Smooth Attention Prior for Multi-Agent Trajectory Prediction
 - Center for Automotive Research at Stanford (CARS) Annual Meeting (2021 – poster)
- Learning Reward Functions from Scale Feedback
 - CoRL 2021 (poster)
- Learning how to Dynamically Route Autonomous Vehicles on Shared Roads
 - ETH Zurich, Institute for Dynamic Systems and Control, Autonomy Talks (2021)
 - The 32nd IEEE Intelligent Vehicles Symposium Workshop (2021)

- 3rd NorCal Control Workshop (2021)
- Stanford Robotics Lunch (2019)
- Center for Automotive Research at Stanford (CARS) Annual Meeting (2020 – poster)
- Walking the Boundary of Learning and Interaction (*with Dorsa Sadigh*)
 - 3rd Robot Learning Workshop: Grounding Machine Learning Development in the Real World @ NeurIPS 2020
- When Humans Aren't Optimal: Robots that Collaborate with Risk-Aware Humans (*with Minae Kwon*)
 - University of Chicago, Graduate Seminar on Topics in Human-Robot Interaction (2020)
- Active Preference-Based Gaussian Process Regression for Reward Learning
 - RSS 2020
- The Green Choice: Learning and Influencing Human Decisions on Shared Roads
 - CDC 2019
- Active Learning of Reward Dynamics from Hierarchical Queries
 - IROS 2019
- Asking Easy Questions: A User-Friendly Approach to Active Reward Learning
 - CoRL 2019 (poster)
- Efficient and Safe Exploration in Deterministic Markov Decision Processes with Unknown Transition Models
 - ACC 2019
 - Stanford AI Safety Retreat 2019 (poster)
- Batch Active Preference-Based Learning of Reward Functions
 - CoRL 2018
 - Stanford HAI 2019 (poster)
 - TRI Joint University Workshop 2019 (poster)
 - Stanford SystemX Fall 2018 (poster)
 - BARS - Bay Area Robotics Symposium (BARS) 2018 (poster)

PROFESSIONAL ACTIVITIES

- **Conferences Organized**
 - **Finance Chair**, Conference on Robot Learning (CoRL), 2024
 - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2021
 - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2020
 - **Organizer**, Emergent Behaviors in Human-Robot Systems (Workshop at Robotics: Science and Systems (RSS)), 2020
 - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2018
- **Program Committee Member**
 - 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2024
 - 7th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2023
- **Editorial Board**

- **Associate Editor** for Robot Learning, IEEE Robotics and Automation Letters (RA-L), 2023 – present
- **Associate Editor**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024
- **Area Chair**, Responsible AI (RAI) (Workshop at International Conference on Learning Representations (ICLR)), 2021
- **Panelist**
 - Human Value Learning Panel at the 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2024
- **Reviewer for Journals**
 - ACM Transactions on Autonomous and Adaptive Systems (TAAS)
 - ACM Transactions on Human-Robot Interaction (THRI)
 - Artificial Intelligence
 - Autonomous Agents and Multi-Agent Systems
 - Autonomous Robots (AURO)
 - Cognitive Systems Research
 - Frontiers in Robotics and AI
 - Nature Communications
 - IEEE Control System Letters (L-CSS)
 - IEEE Robotics and Automation Letters (RA-L)
 - IEEE Robotics and Automation Magazine (RAM)
 - IEEE Transactions on Automatic Control (TAC)
 - IEEE Transactions on Control Systems Technology (TCST)
 - IEEE Transactions on Human-Machine Systems (THMS)
 - IEEE Transactions on Intelligent Transportation Systems (T-ITS)
 - IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
 - IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
 - IEEE Transactions on Robotics (T-RO)
 - Journal of Machine Learning Research (JMLR)
 - Proceedings of the Royal Society A
 - Robotics
 - The International Journal of Robotics Research (IJRR)
- **Reviewer for Conferences**
 - Conference on Neural Information Processing Systems (NeurIPS), 2023–2024
 - Conference on Robot Learning (CoRL), 2019–2024
 - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020, 2022–2024
 - IEEE International Conference on Robot & Human Interactive Communication (RO-MAN), 2024
 - Robotics: Science and Systems (RSS), 2020–2021, 2023–2024
 - Conference on Uncertainty in Artificial Intelligence (UAI), 2023–2024
 - Learning for Dynamics & Control (L4DC), 2020, 2023–2024
 - IEEE International Conference on Robotics and Automation (ICRA), 2021–2024
 - ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2020–2024

- IEEE Conference on Decision and Control (CDC), 2019–2020, 2023–2024
- International Joint Conference on Artificial Intelligence (IJCAI), 2023
- American Control Conference (ACC), 2019, 2021–2022
- IEEE Conference on Control Technology and Applications (CCTA), 2021
- IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2020
- International Conference on Computer-Aided Verification (CAV), 2019
- ACM International Conference on Hybrid Systems: Computation and Control (HSCC), 2019

- **Reviewer for Workshops**

- RSS Pioneers Workshop 2023–2024
- HRI Pioneers Workshop 2023–2024
- Interactive Learning with Implicit Human Feedback Workshop (ILHF) at ICML 2023
- AAMAS Workshop on Rebellious and Disobedient AI (RaD-AI), 2022–2023
- CoRL Workshop on Aligning Robot Representations with Humans, 2022
- NeurIPS Workshop on Foundation Models for Decision Making (FMDM), 2022
- NeurIPS Workshop on Progress and Challenges in Building Trustworthy Embodied AI (TEA), 2022
- NeurIPS Workshop on Machine Learning Safety, 2022
- RSS Workshop on Learning from Diverse, Offline Data (L-DOD), 2022
- RSS Workshop on Social Intelligence in Humans and Robots, 2022
- Artificial Intelligence for Human-Robot Interaction Symposium (AI-HRI) at AAAI Fall Symposium Series, 2021
- ICRA Workshop on Social Intelligence in Humans and Robots, 2021
- Bridging AI and Cognitive Science (BAICS) (Workshop at International Conference on Learning Representations (ICLR)), 2020

OUTREACH AND MENTORING

- **Berkeley Artificial Intelligence Safety Initiative for Students, 2023-present** Berkeley, CA, USA
 - **Mentor at Supervised Program for Alignment Research:** Mentoring three undergraduate students on a multi-agent inverse reinforcement learning research project
- **Berkeley Artificial Intelligence Research Lab (BAIR), 2022-2023** Berkeley, CA, USA
 - **BAIR Undergraduate Mentoring Program Mentor:** Mentoring undergraduate students about their future career, getting involved in AI research, and applying for industry or graduate school
- **Stanford University Department of Computer Science, 2020-2022** Stanford, CA, USA
 - **CS Mentorship Program Organizer:** Organizing the CS mentorship program and the related social events
 - **CS Mentorship Program Mentor:** Mentoring first and second-year undergraduate students about their future career, getting involved in research projects, and applying for industry or graduate school
- **Stanford University AI Laboratory (SAIL), 2018-2020** Stanford, CA, USA
 - **AI Mentorship Program Organizer:** Organizing the AI mentorship program and the related social events
 - **AI Mentorship Program Mentor:** Mentoring first and second-year undergraduate students about their future career, getting involved in AI research, and applying for industry or graduate school
 - **Robotics Lunch Organizer (2019-2020):** Organizing bi-weekly robotics lunch sessions where invited professors, postdoctoral researchers and Ph.D. candidates present their research
- **Stanford STEM to SHTeM Summer Internship Program, 2019** Stanford, CA, USA

- **Mentor:** Mentoring high school students in a research project about decision making under risk / time constraints
- **IEEE Bilkent Student Branch, 2012-2017** Ankara, Turkey
 - **Road to University Volunteer (2013-2017):** Introducing engineering and campus life to high school students from all around Turkey
 - **Vice Chair (2014-2015):** Managing and organizing the technical trainings and social events
 - **Robotics and Automation Society Member (2012-2015):** Assistantship in the electronics and robotics focused tutorials
 - **Web Team Member (2012-2014) and Webmaster (2013-2014):** Designing and managing the website of IEEE Bilkent SB, its communities and events; teaching in web design tutorials and in MATLAB tutorials; organizing a Java programming competition and weekly brain teasers
- **GazeteBilkent, 2014-2017** Ankara, Turkey
 - **Online Operations Manager:** Managing the website of GazeteBilkent, an online newspaper
- **Bilkent University, 2014-2017** Ankara, Turkey
 - **Webmaster of Electrical and Electronics Engineering (2013-2017), Mechanical Engineering (2015-2017), Economics (2016-2017) Departments:** Designing, developing and managing the websites of the departments and developing necessary web tools
 - **1st and 2nd Industrial Design Projects Fairs Student Coordinator and Webmaster (2015, 2016):** Organizing the fair in which the senior students of Bilkent EEE present their industry projects
 - **Graduate Research Conference '15 Student Coordinator (2015):** Organizing the conference series in which graduate students of Bilkent EEE present their research and projects
 - **Graduate Research Conference '14 Webmaster and Organization Team Member (2014):** Organizing the conference series in which graduate students of Bilkent EEE present their research and projects
- **Bilkent Chess Society, 2013-2014** Ankara, Turkey
 - **Vice Chair:** Organizing tournaments on campus, participating in local events as a team
- **Bilkent Academic Career Club, 2013** Ankara, Turkey
 - **Physics Olympiads Volunteer:** Organizing a physics competition among high school students

PATENTS

1. Z Cao, **E Bıyık**, WZ Wang, A Raventos, A Gaidon, G Rosman, D Sadigh. “Reinforcement Learning Based Control of Imitative Policies for Autonomous Driving”, US Patent Application No. US17/002,650.

JOURNAL PUBLICATIONS

1. **E Bıyık**, N Anari, D Sadigh. “Batch Active Learning of Reward Functions from Human Preferences”, ACM Transactions on Human-Robot Interaction (THRI), 2024.
2. S Casper*, X Davies*, C Shi, TK Gilbert, J Scheurer, J Rando, R Freedman, T Korbak, D Lindner, P Freire, T Wang, S Marks, CR Segerie, M Carroll, A Peng, P Christoffersen, M Damani, S Slocum, U Anwar, A Siththaranjan, M Nadeau, EJ Michaud, J Pfau, D Krashennnikov, X Chen, L Langosco, P Hase, **E Bıyık**, A Dragan, D Krueger, D Sadigh, D Hadfield-Menell (*equal contribution). “Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback”, Transactions on Machine Learning Research (TMLR), 2023.
3. **E Bıyık**, N Huynh, MJ Kochenderfer, D Sadigh. “Active Preference-Based Gaussian Process Regression for Reward Learning and Optimization”, The International Journal of Robotics Research (IJRR), 2023; doi: 10.1177/02783649231208729.
4. M Tucker, K Li, E Novoseller, M Pétriaux, G Burger, **E Bıyık**, M Masselin, D Sadigh, JW Burdick, Y Yue, AD Ames. *Anonymous Submission*, Nature Machine Intelligence, 2022. (*Submitted*)
5. **E Bıyık**, DP Losey, M Palan, NC Landolfi, G Shevchuk, D Sadigh. “Learning Reward Functions from Diverse Sources of Human Feedback: Optimally Integrating Demonstrations and Preferences”, The International Journal of Robotics Research (IJRR), 2022; doi:10.1177/02783649211041652

6. DA Lazar*, **E Biyik***, D Sadigh, R Pedarsani (*equal contribution). “Learning How to Dynamically Route Autonomous Vehicles on Shared Roads”, *Transportation Research Part C: Emerging Technologies (TR.C)*, 2021; doi: 10.1016/j.trc.2021.103258
7. **E Biyik***, DA Lazar*, R Pedarsani, D Sadigh (*equal contribution). “Incentivizing Efficient Equilibria in Traffic Networks with Mixed Autonomy”, *IEEE Transactions on Control of Network Systems (TCNS)*, 2021; doi: 10.1109/TCNS.2021.3084045.
8. **E Biyik***, K Keskin*, SUH Dar, A Koc, T Çukur (*equal contribution). “Factorized sensitivity estimation for artifact suppression in phase-cycled bSSFP MRI”, *NMR in Biomedicine*, 2020; doi: 10.1002/nbm.4228.
9. **E Biyik**, E Ilicak, T Çukur. “Reconstruction by Calibration over Tensors for Multi-Coil Multi-Acquisition Balanced SSFP Imaging”, *Magnetic Resonance in Medicine (MRM)*, 2017; doi: 10.1002/mrm.26902.
10. E Ilicak, LK Senel, **E Biyik**, T Çukur. “Profile-encoding reconstruction for multiple-acquisition balanced steady-state free precession imaging”, *Magnetic Resonance in Medicine (MRM)*, 2016; doi: 10.1002/mrm.26507.

REFEREED CONFERENCE PUBLICATIONS

11. Y Wang*, Z Sun*, J Zhang, Z Xian, **E Biyik**, D Held[†], Z Erickson[†] (*equal contribution, [†]equal advising). “RL-VLM-F: Reinforcement Learning from Vision Language Foundation Model Feedback”, 41st International Conference on Machine Learning (ICML), Vienna, Austria, Jul. 2024.
12. M Pan*, ML Schrum*, V Myers, **E Biyik**, A Dragan (*equal contribution). *Coprocessor Actor Critic: A Model-Based Reinforcement Learning Approach For Adaptive Brain Stimulation*, 41st International Conference on Machine Learning (ICML), Vienna, Austria, Jul. 2024.
13. E Ellis, GR Ghosal, SJ Russell, A Dragan, **E Biyik**. *A Generalized Acquisition Function for Preference-based Reward Learning*, 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.
14. S Sontakke, J Zhang, S Arnold, K Pertsch, **E Biyik**, D Sadigh, C Finn, L Itti. *RoboCLIP: One Demonstration is Enough to Learn Robot Policies*, 37th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Louisiana, USA, Nov. 2023.
15. V Myers, **E Biyik**, D Sadigh. *Asking Preference Questions Online in Active Reward Learning*, 2023 IEEE International Conference on Robotics and Automation (ICRA), London, United Kingdom, May 2023.
16. M Srivastava, **E Biyik**, S Mirchandani, ND Goodman, D Sadigh. *Assistive Teaching of Motor Control Tasks to Humans*, 36th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Louisiana, USA, Nov. 2022.
17. E Brockbank, H Wang, J Yang, S Mirchandani, **E Biyik**, D Sadigh, J Fan. *How do People Incorporate Advice from Artificial Agents when Making Physical Judgments?*, 44th Annual Meeting of the Cognitive Science Society (CogSci), Toronto, Ontario, Canada, Jul. 2022.
18. Z Cao, **E Biyik**, G Rosman, D Sadigh. “Leveraging Smooth Attention Prior for Multi-Agent Trajectory Prediction”, 2022 IEEE International Conference on Robotics and Automation (ICRA), Philadelphia, Pennsylvania, USA, May 2022; doi: 10.1109/ICRA46639.2022.9811718.
19. **E Biyik**, A Talati, D Sadigh. *APReL: A Library for Active Preference-based Reward Learning Algorithms*, ACM/IEEE International Conference on Human-Robot Interaction (HRI), Sapporo, Hokkaido, Japan, Mar. 2022.
20. **E Biyik**, A Lalitha, R Saha, A Goldsmith, D Sadigh. *Partner-Aware Algorithms in Decentralized Cooperative Bandit Teams*, Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI), Vancouver, British Columbia, Canada, Feb. 2022; doi: 10.1609/aaai.v36i9.21158.
21. N Wilde*, **E Biyik***, D Sadigh, SL Smith (*equal contribution). *Learning Reward Functions from Scale Feedback*, 5th Conference on Robot Learning (CoRL), London, United Kingdom, Nov. 2021.
22. V Myers, **E Biyik**, N Anari, D Sadigh. *Learning Multimodal Rewards from Rankings*, 5th Conference on Robot Learning (CoRL), London, United Kingdom, Nov. 2021.
23. WZ Wang*, M Beliaev*, **E Biyik***, DA Lazar, R Pedarsani, D Sadigh (*equal contribution). *Emergent Prosociality in Multi-Agent Games Through Gifting*, 30th International Joint Conference on Artificial Intelligence (IJCAI), Montreal, Quebec, Canada, Aug. 2021; doi: 10.24963/ijcai.2021/61.
24. K Li, M Tucker, **E Biyik**, E Novoseller, JW Burdick, Y Sui, D Sadigh, Y Yue, AD Ames. “ROIAL: Region of Interest Active Learning for Characterizing Exoskeleton Gait Preference Landscapes”, 2021 IEEE International Conference on Robotics and Automation (ICRA), Xi’an, China, May 2021; doi: 10.1109/ICRA48506.2021.9560840.
25. M Beliaev, **E Biyik**, DA Lazar, WZ Wang, D Sadigh, R Pedarsani. “Incentivizing Routing Choices for Safe and Efficient Transportation in the Face of the COVID-19 Pandemic”, 12th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS), Nashville, Tennessee, USA, May 2021; doi: 10.1145/3450267.3450546.

26. Z Zhu, **E Bıyık**, D Sadigh. “Multi-Agent Safe Planning with Gaussian Processes”, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Las Vegas, Nevada, USA, Oct. 2020; doi: 10.1109/IROS45743.2020.9341169.
27. **E Bıyık***, N Huynh*, MJ Kochenderfer, D Sadigh (*equal contribution). “Active Preference-Based Gaussian Process Regression for Reward Learning”, Robotics: Science and Systems (RSS), Corvallis, Oregon, USA, Jul. 2020; doi: 10.15607/rss.2020.xvi.041.
28. Z Cao*, **E Bıyık***, WZ Wang, A Raventos, A Gaidon, G Rosman, D Sadigh (*equal contribution). “Reinforcement Learning based Control of Imitative Policies for Near-Accident Driving”, Robotics: Science and Systems (RSS), Corvallis, Oregon, USA, Jul. 2020; doi: 10.15607/rss.2020.xvi.039
29. M Kwon, **E Bıyık**, A Talati, K Bhasin, DP Losey, D Sadigh. “When Humans Aren’t Optimal: Robots that Collaborate with Risk-Aware Humans”, ACM/IEEE International Conference on Human-Robot Interaction (HRI), Cambridge, United Kingdom, Mar. 2020; doi: 10.1145/3319502.3374832. *Honorable mention award*.
30. **E Bıyık**, DA Lazar, D Sadigh, R Pedarsani. “The Green Choice: Learning and Influencing Human Decisions on Shared Roads”, 58th IEEE Conference on Decision and Control (CDC), Nice, France, Dec. 2019; doi: 10.1109/CDC40024.2019.9030169.
31. C Basu, **E Bıyık**, Z He, M Singhal, D Sadigh. “Active Learning of Reward Dynamics with Hierarchical Queries”, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Macau, China, Nov. 2019; doi: 10.1109/IROS40897.2019.8968522.
32. **E Bıyık**, M Palan, NC Landolfi, DP Losey, D Sadigh. “Asking Easy Questions: A User-Friendly Approach to Active Reward Learning”, 3rd Conference on Robot Learning (CoRL), Osaka, Japan, Oct. 2019.
33. **E Bıyık***, J Margoliash*, SR Alimo, D Sadigh (*equal contribution). “Efficient and Safe Exploration in Deterministic Markov Decision Processes with Unknown Transition Models”, American Control Conference (ACC), Philadelphia, Pennsylvania, USA, Jul. 2019; doi: 10.23919/ACC.2019.8815276.
34. **E Bıyık***, DA Lazar*, R Pedarsani, D Sadigh (*equal contribution). “Altruistic Autonomy: Beating Congestion in Shared Roads”, Workshop on Algorithmic Foundations of Robotics (WAFR), Mérida, México, Dec. 2018; doi: 10.1007/978-3-030-44051-0_51
35. **E Bıyık**, D Sadigh. “Batch Active Preference-Based Learning of Reward Functions”, 2nd Conference on Robot Learning (CoRL), Zürich, Switzerland, Oct. 2018.
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